

MOCK INTERVIEW – FINAL REPORT

Course: Computer Networks | **Course Code:** CSA07 | **Assessment:** Tool 3 – Mock Interview

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Q1

Question: What is DNS, and why is it important in a computer network?

Response: DNS (Domain Name System) is the digital phonebook of the internet. It translates human-friendly website names into numeric IP addresses that computers use to talk to each other. Why it is important: Easy to use: You only have to remember names like google.com instead of long numbers like 142.250.190.46. Connects devices: Computers and networks cannot understand text names; they require the numeric IP address to load a website or send an email. Without DNS, you would have to memorize a long string of numbers for every single website you want to visit.

Q2

Question: What is the difference between SMTP and POP, and what is the role of each in electronic mail communication?

Response: SMTP (Simple Mail Transfer Protocol) and POP (Post Office Protocol) serve opposite roles to make email work: SMTP acts as the sender, while POP acts as the retriever. When you compose an email and hit send, SMTP takes that message from your device and routes it across the internet to the destination server. Once the message arrives at the destination, POP steps in on the recipient's end to connect to that server, download the email onto a single local device, and typically clear it off the server. Put simply, SMTP is the postal carrier delivering your mail to the post office, and POP is you picking that mail up out of your physical P.O. box to take home.

Q3

Question: What is P2P (Peer-to-Peer) communication, and how is it different from the traditional client-server model?

Response: Peer-to-Peer (P2P) is a network structure where computers connect and share data directly with one another without needing a central middleman or main server. In a P2P setup, every computer (called a "peer") acts as both a customer requesting data and a provider sharing data.

Q4

Question: What is VoIP, and how does it allow people to communicate over a computer network?

Response: VoIP stands for Voice over Internet Protocol, which simply means making phone calls using the internet instead of traditional phone lines. Whenever you talk to someone using apps like Discord, WhatsApp, Skype, or Zoom, you are using VoIP technology.

Q5

Question: What is SSL, and how does it help protect data when communicating over a network?

Response: SSL (Secure Sockets Layer) is a security technology that creates an encrypted link between a web browser and a website server. It protects your data using three main steps: Encryption: It scrambles sensitive data (like passwords, credit card numbers, and personal details) into unreadable code before it travels over the network. Authentication: It uses digital certificates to verify that you are connecting to the real website rather than a fake site set up by an attacker. Data Integrity: It ensures the data cannot be altered, tampered with, or corrupted during transit without being detected.

Q6

Question: What is a firewall? How does it protect a network from unauthorized access and possible attacks?

Response: A firewall is a cybersecurity system that acts as a protective barrier between a private network (like your home Wi-Fi or computer) and public networks (like the internet). Think of it as a security guard standing at the front door of a building, checking everyone's ID before letting them in or out.

Q7

Question: Imagine that users in a company can access some websites but cannot open a particular website. What possible network-related causes would you check, and how would you troubleshoot the problem?

Response: When users can reach most websites but fail to access one specific site, the issue is almost always local to DNS, routing, firewall rules, or the target server itself.

Q8

Question: A company is experiencing a DoS (Denial-of-Service) attack that is making its website very slow or unavailable. What security mechanisms can be used to reduce the impact of the attack?

Response: Defending against a Denial-of-Service (DoS) or Distributed Denial-of-Service (DDoS) attack requires a layered defense strategy to filter malicious requests, distribute traffic load, and preserve access for legitimate users.

Q9

Question: A company wants to provide voice communication between employees in different locations and also allow them to share files directly without depending completely on a central server. Which technologies or communication approaches would you choose, and why?

Response: To meet these requirements, the best architectural approach is a Hybrid Peer-to-Peer (Hybrid P2P) Network model combining real-time communication protocols like WebRTC and VoIP. Instead of routing every piece of voice audio and large document through a central server, employees connect directly to one another computer-to-computer.

Core Technologies: WebRTC (Web Real-Time Communication) allows two devices to open a direct, encrypted connection. It handles live voice audio, video, and direct file transfers via WebRTC DataChannels. VoIP (Voice over IP) / SIP is used for voice communication.

Why choose this approach? Direct data transfer can reduce central-server bottlenecks; direct communication can reduce latency; it can improve scalability and reliability. A light signaling server can still assist with discovery and handshake before the peers establish a direct encrypted connection.

Q10

Question: A company wants to securely provide employees with access to its internal services over the internet. Which combination of network protocols and security mechanisms would you recommend, and why?

Response: To grant employees secure remote access to internal services over the public internet, the ideal architecture relies on a WireGuard/IPsec VPN or a Zero Trust Network Access (ZTNA) model paired with strong identity controls.

Question-wise Evaluation

Question	Level	SubQ Total
Q1	Level 4	5.0
Q2	Level 4	5.0
Q3	Level 3	3.75
Q4	Level 3	3.75
Q5	Level 4	5.0
Q6	Level 3	3.75
Q7	Level 2	2.5
Q8	Level 3	3.75
Q9	Level 4	5.0
Q10	Level 3	3.75

Final Rubric Evaluation

Rubric	Maximum	Evaluation
Technical Knowledge	25	Strong understanding of core networking concepts and security mechanisms.
Problem Solving	25	Good application of networking concepts, especially in Q9 and Q10.
Analytical Thinking	20	Good scenario-based reasoning, with Q7 needing more troubleshooting detail.
Communication	15	Clear explanations and useful examples/analogies.
Confidence	15	Responses were direct and technically focused.
Total	100	Good overall performance

Student Avg Final Mark (Q1–Q10 SubQ Totals): 4.12 / 5.00

Overall Feedback

Strengths: Good understanding of DNS, email protocols, P2P, VoIP, SSL, firewalls, DoS/DDoS protection, and hybrid communication architecture. The use of practical examples and analogies made several answers easy to understand.

Areas for Improvement: Some answers could explain the technical process more fully. In particular, Q6 could mention packet filtering/rules, and Q7 could give a clearer step-by-step troubleshooting sequence. Q10 could explain why VPN or ZTNA is selected and how identity controls strengthen security.

Interview Readiness: Good. The student demonstrates a solid foundation in computer networking and can explain concepts in understandable language. More structured troubleshooting and deeper justification would improve advanced interview performance.